

Baryon Octets

$$3^F \times 3^F \times 3^F = 1^F_A + 8^F_{12} + 8^F_{23} + 10^F_S$$

$$\psi_{\text{Baryon}} = \psi_{\text{space}} \underbrace{\psi_{\text{spin}} \psi_{\text{flav}}}_{\text{S}} \psi_{\text{col.}}$$

A S A

$$B = 999$$

Baryon is a fermion.

ground state $L=0$

symm is $(-1)^L$ for ψ_{space}

$$1^F \text{ cannot exist} \quad 10^F \quad S=1/2 \text{ cannot exist}$$

ψ_{Baryon} argum. under exchange of $1 \leftrightarrow 2, 2 \leftrightarrow 3, 1 \leftrightarrow 3$
for spin, flav. and color

$$8^F_{12} + 8^F_{23} \text{ partially symm.}$$

For ψ_{Baryon}^8 to exist $\Rightarrow \psi_{\text{spin}} \psi_{\text{flav}} = \text{always symm}$
under any exchange.

octet with $S=3/2$ $\uparrow\uparrow\uparrow, \downarrow\downarrow\downarrow$ always symm.

$$\psi_{12}^F \psi^{S=3/2} + \psi_{23}^F \psi^{S=3/2}$$

$\Rightarrow$ there are some combinations that are not symm.

$\Rightarrow$ no octet with $S=3/2$

Baryon octet with $S=1/2$

$\uparrow\uparrow\downarrow$
 S_{12}

$\uparrow\downarrow\uparrow$
 S_{13}

$\downarrow\uparrow\uparrow$
 S_{23}

$$\frac{1}{2} \times \frac{1}{2} \times \frac{1}{2} = \left(\frac{3}{2}\right)_S + \left(\frac{1}{2}\right)_{12} + \left(\frac{1}{2}\right)_{23}$$

From Clebsch-Gordan coeff

$$\left(\frac{1}{2}\right)_{12} + \left(\frac{1}{2}\right)_{23} = \left(\frac{1}{2}\right)_{13}$$

$$\psi_{\text{spin}} \psi_{\text{flav}} = \underbrace{\psi_{12}^{S=1/2} \psi_{F,12}^{F=8} + \psi_{23}^{S=1/2} \psi_{F,23}^{F=8}}_{\text{can be obtained}} + \psi_{13}^{S=1/2} \psi_{F,13}^{F=8}$$

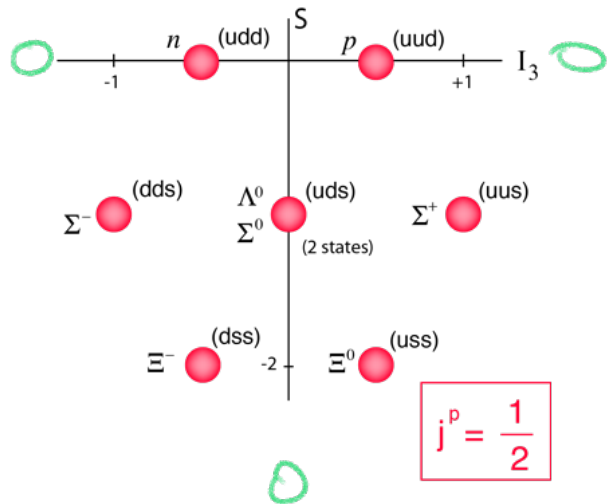
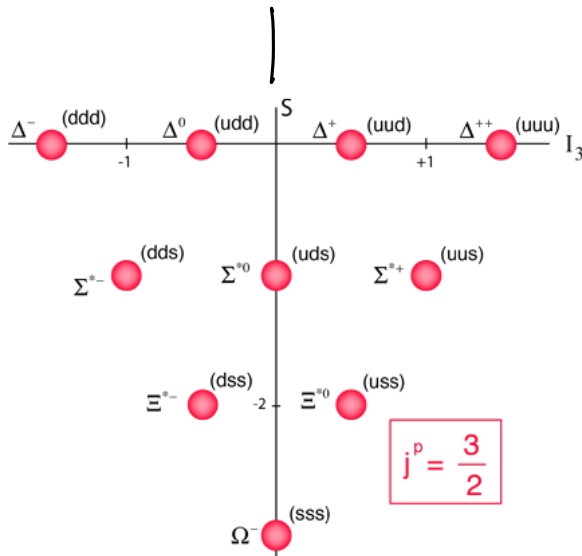
Flavor octet with $S=1/2$ allowed from symm.

Baryons could have 27 states $\Rightarrow$ 10 + 8 physical states.

Satisfy or symm requirements.

10 with $S=3/2$.

Octet with $S=1/2$

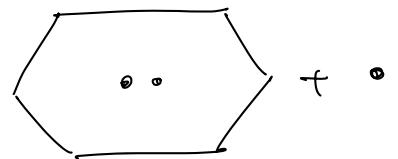


what about mesons?

$$M = q_1 \bar{q}_2$$

$$3_F \times 3_F = 14_{F_1 A} + 8_{F_1 S}$$

$$3_C \times 3_C = 14_{C_1 A} + 8_{C_1 S}$$



$$\psi_{\text{meson}} = \psi_{\text{space}} \psi_{\text{spin}} \psi_{\text{flavor}} \psi_{\text{color}}$$

$$\pi^+ = u \bar{d} \quad 1 \xrightarrow{F} 2 \quad d \bar{u}$$

$q_1 \neq \bar{q}_2 \Rightarrow$ No symm. constraint because not identical particles.

$$\text{spin for mesons: } \frac{1}{2} \times \frac{1}{2} = 0 + 1$$

$\Rightarrow$ all combinations allowed with $S=0, S=1$

$\Rightarrow$ physical evidence supports existence of color

$\Rightarrow$ direct evidence for color and $\neq$ color

$$p + p \rightarrow p + p + \pi^+ \pi^-$$

hadron scattering not optimal to test color.

leptons, or photons have no color

e^+e^- collisions.

$$\begin{array}{ccc}
 e^+ & e^- & \longrightarrow q_1 \bar{q}_2 \\
 B & 0 & 0 \\
 Q & +1 & -1 \qquad +\frac{2}{3} \quad -\frac{2}{3} \\
 & & -\frac{1}{3} \quad +\frac{1}{3}
 \end{array}$$

Color has to be conserved in QED

$$\begin{array}{ccc}
 \text{color} & 0 & 0 \\
 & R & \bar{R} \\
 & B & \bar{B} \\
 & G & \bar{G}
 \end{array}$$

$$e^+ e^- \longrightarrow u_R \bar{u}_R$$

$$\cancel{u \bar{d}} \quad Q \neq 0$$

$$\begin{array}{ccc}
 e^+ & e^- & \longrightarrow u_G \bar{c}_{\bar{G}} \\
 Q & +\frac{2}{3} & -\frac{2}{3}
 \end{array}$$

$\Delta C = 1$
Charm quantum number
not conserved.

QED conserves color and Flavor.

$$e^+ e^- \longrightarrow c_B \bar{c}_{\bar{B}}$$

$$\begin{pmatrix} u \\ d \end{pmatrix} \begin{pmatrix} c \\ s \end{pmatrix} \begin{pmatrix} t \\ b \end{pmatrix} \quad \begin{array}{l} Q = +2/3 \\ Q = -1/3 \end{array}$$

$SU(3)_F \Rightarrow$ takes into account u, d, s .

$$\begin{pmatrix} u \\ d \\ s \end{pmatrix}_{\text{Flavor}}$$

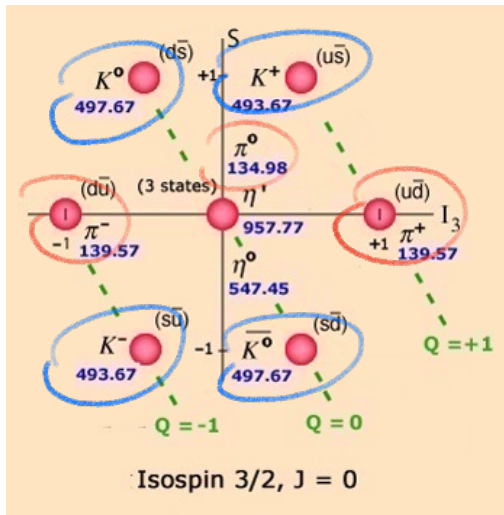
What about c, b, t ?

~~SU(3)~~ $SU(3)_F$ conserved $\Rightarrow$ mesons should have same mass in octets.

Baryons should have the same mass in 40, 8

Symm $\Rightarrow$ I can exchange $u \leftrightarrow d \leftrightarrow s$ w/o consequence

$\Rightarrow$ assuming $m_u \approx m_d \approx m_s$



$$m_K \neq m_\pi \Rightarrow m_S \neq (m_u \approx m_d)$$

mass pattern $\Rightarrow$

$SU(3)_F$ is approximate symm
because $m_S \neq (m_d, m_u)$

Baryons	qqq	J^P	I	I_3	S	$Q^{(1)}$	mass (MeV)
p, n	uud, udd	$\frac{1}{2}^+$	$\frac{1}{2}$	$\frac{1}{2}, -\frac{1}{2}$	0	1, 0	940
Λ	uds	$\frac{1}{2}^+$	0	0	-1	0	1115
$\Sigma^+, \Sigma^0, \Sigma^-$	uus, uds, dds	$\frac{1}{2}^+$	1	1, 0, -1	-1	1, 0, -1	1190
Ξ^0, Ξ^-	uss, dss	$\frac{1}{2}^+$	$\frac{1}{2}$	$\frac{1}{2}, -\frac{1}{2}$	-2	1, 0	1320
$\Delta^{++}, \Delta^+, \Delta^0, \Delta^-$	uuu, uud, udd, ddd	$\frac{3}{2}^+$	$\frac{3}{2}$	$\frac{3}{2}, \frac{1}{2}, -\frac{1}{2}, -\frac{3}{2}$	0	2, 1, 0, -1	1230
$\Sigma^{*+}, \Sigma^{*0}, \Sigma^{*-}$	uus, uds, dds	$\frac{3}{2}^+$	1	1, 0, -1	-1	1, 0, -1	1385
Ξ^{*0}, Ξ^{*-}	uss, dss	$\frac{3}{2}^+$	$\frac{1}{2}$	$\frac{1}{2}, -\frac{1}{2}$	-2	1, 0	1530
Ω^-	sss	$\frac{3}{2}^+$	0	0	-3	-1	1670

Expenim. do not measure mass of quarks.

only mass of baryons and mesons.

$\Rightarrow$ extrapolate nominal mass for quarks

$$SU(3)_F: \quad u = \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix} \quad d = \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix} \quad s = \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix} \quad \begin{array}{c} d \quad u \\ | \\ s \end{array} \rightarrow I_3$$

$I, \text{ Strangeness.}$

Assume to have $m_u \approx m_d \approx m_s \approx m_c$

$$u = \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix} \quad d = \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix} \quad s = \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix} \quad c = \begin{pmatrix} 0 \\ 0 \\ 0 \\ 1 \end{pmatrix}$$

$SU(4)_{\text{Flavor}}$

$N^2 - 1 = 15$ generators

$N - 1 = 3$ diagonal generators.

Isospin, Strangeness, Charm

~~and~~ $\begin{pmatrix} u \\ d \end{pmatrix} = I=1/2$ doublet. S, C are Isospin Singlet

$S=0$ for u, d, C . $S=-1$ for strange quark

$C=0$ for u, d, S $C=1$ for charm quark.

$$Q = I_3 + \frac{B+S+C}{2}$$

$\begin{pmatrix} u \\ d \\ s \\ c \end{pmatrix}$ or fundam. rep. of $SU(4)$

$\begin{pmatrix} \bar{u} \\ \bar{d} \\ \bar{s} \\ \bar{c} \end{pmatrix}$ are conjugate fundam. rep

$$q \times \bar{q} = 4 \times \bar{4} = 14 + 15 = 1 + 8 + 6 + 1 \quad \begin{matrix} \nearrow C=0 \\ \downarrow C \neq 0 \end{matrix}$$

$\underbrace{C=0}_{\text{mesons w/o } C}$

Charmed Mesons:

up-like $\rightarrow \begin{pmatrix} u \\ d \end{pmatrix} \begin{pmatrix} C \\ S \end{pmatrix} \begin{pmatrix} t \\ b \end{pmatrix} \quad \begin{matrix} Q = +2/3 \\ Q = -1/3 \end{matrix}$

down-like $\rightarrow$

$c \bar{u} \quad D^0$

$c \bar{d} \quad D^+$

$c \bar{s} \quad D_s^+ \quad (D \text{ sub } S)$

$\left. \begin{matrix} \bar{c} u \\ \bar{c} d \\ \bar{c} s \end{matrix} \right\} C = -1$

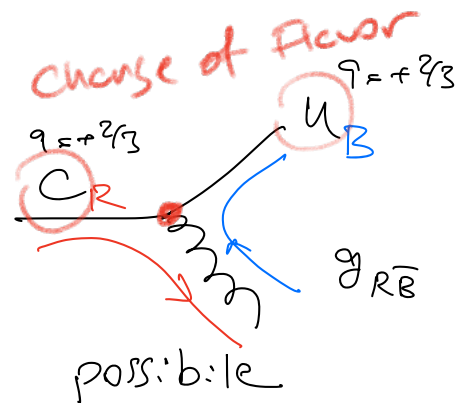
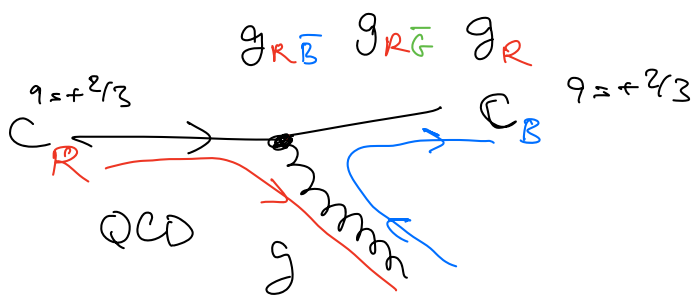
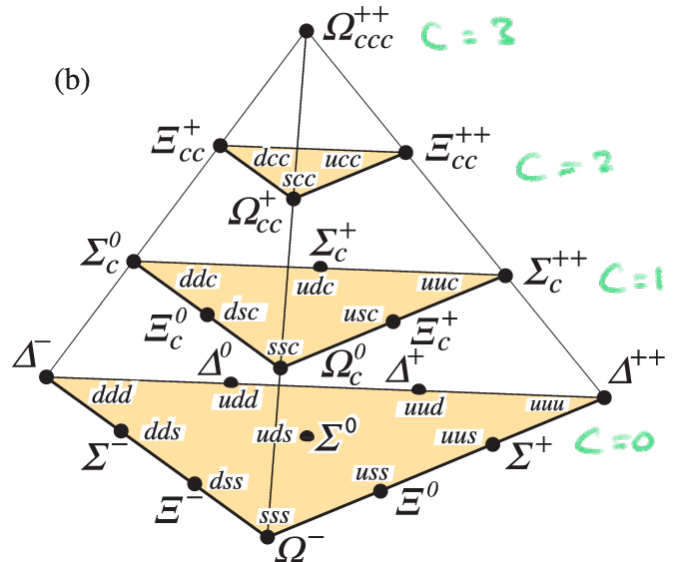
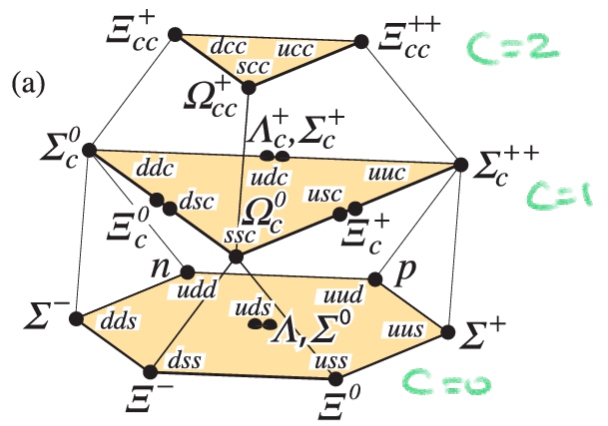
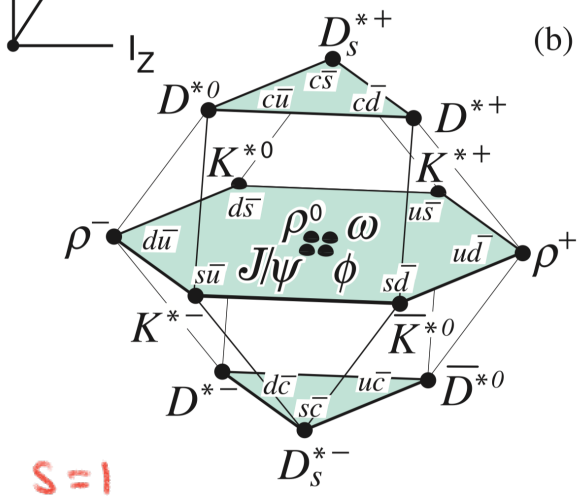
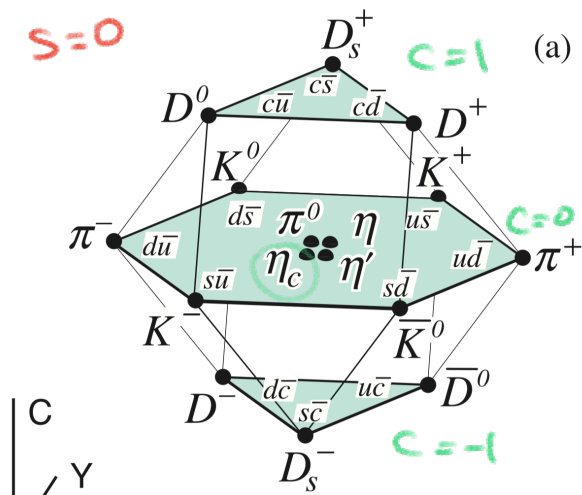
$J/\psi = c \bar{c} \quad S=0$

$\Upsilon/\psi = c \bar{c} \quad S=1$

$m_{D^0, D^+} \approx 1.864 \text{ GeV} \gg m_K = 498 \text{ MeV}$
 $\gg m_\pi = 140 \text{ MeV}$

$\Rightarrow m_C \gg m_S \gg m_{u,d}$

$\Rightarrow SU(4)$ Broken.



$R\bar{G}, R\bar{B}, G\bar{B}, G\bar{R}, R\bar{B}, R\bar{G}, \frac{1}{\sqrt{2}}(R\bar{R} - G\bar{G}), \frac{1}{\sqrt{6}}(\frac{2}{3}R\bar{R} - \frac{1}{3}G\bar{G} - 2B\bar{B}) \Rightarrow \Delta C \neq 0$
in QCD.

Assumption of QED, QCD conserve color and flavor.

(No experimental evidence of violation so far)

b: 5th quark $\Rightarrow$ $SU(5) \Rightarrow$ 4 diagonal generators.
 $\Rightarrow$ Isospin, Strangeness, charm, beauty.

Gell-Mann
Nishijima

$$Q = I_3 + \frac{B+S+C+b}{2}$$

B mesons: $b\bar{u}$ B^- $b\bar{d}$ B^0 $\bar{b}u$ B^+

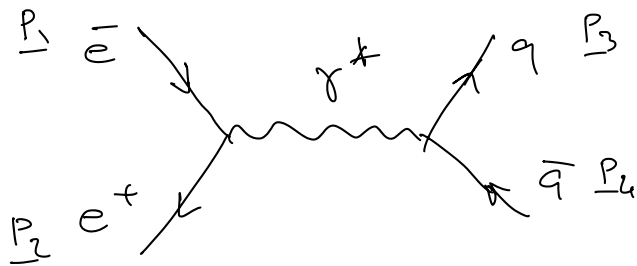
mass of B mesons $\sim 5.279 \text{ GeV} \gg m_D \sim 2 \text{ GeV} \gg$

$\gg m_K = 0.5 \gg m_\pi \sim 0.14 \text{ GeV}$

$$\begin{pmatrix} u \\ d \end{pmatrix} \begin{pmatrix} c \\ s \end{pmatrix} \begin{pmatrix} t \\ b \end{pmatrix}$$

Direct evidence for quarks, and color $\Rightarrow e^+e^-$ collisions. through QED.

$$e^+e^- \rightarrow q\bar{q}$$



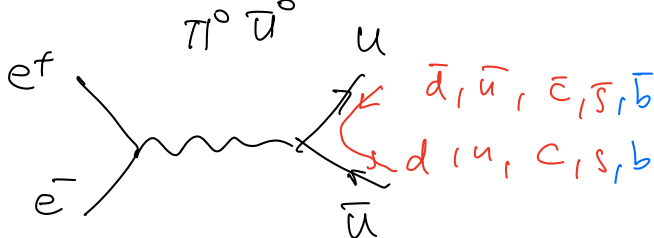
S-channel or annihilation diagram.

$$\sqrt{s} \geq 2m_q$$

$$s = (\underline{P}_1 + \underline{P}_2)^2 = (\underline{P}_3 + \underline{P}_4)^2$$

if quarks are not free $\Rightarrow$ only mesons and baryons can be produced.

$$e^+e^- \rightarrow \pi^+\pi^-$$



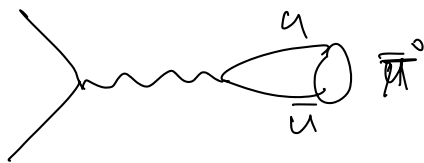
$$\begin{array}{llll} \pi^+ & \bar{u} & D^+ & K^+ & B^+ & B^0 \\ \pi^- & u & D^- & K^- & B^- & \bar{B}^0 \end{array}$$

$$\sqrt{s} \geq 2m_\pi$$

$$\text{for } \pi^+\pi^-, \pi^0\pi^0 = 280 \text{ MeV}$$

$$\geq 2m_D$$

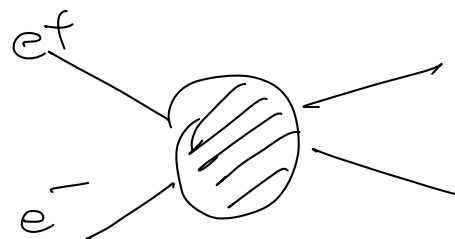
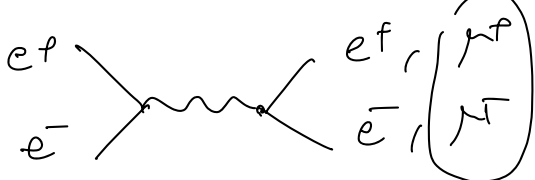
$$\text{for } D^+D^-$$



$$e^+e^- \rightarrow \pi^0$$

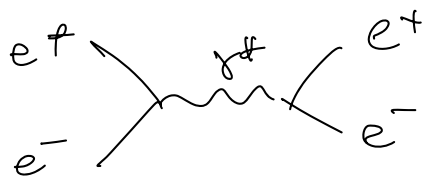
$$\pi^0 \rightarrow e^+e^- ?$$

$$\sqrt{s} < 2m_\pi = 280 \text{ MeV}$$

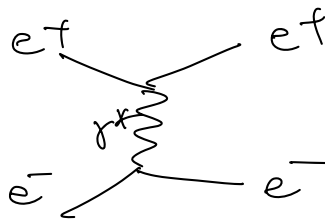


$$\sqrt{s} \geq 2m_\mu = 212 \text{ MeV.}$$

$$\sqrt{s} < 200 \text{ MeV}$$



s-Channel



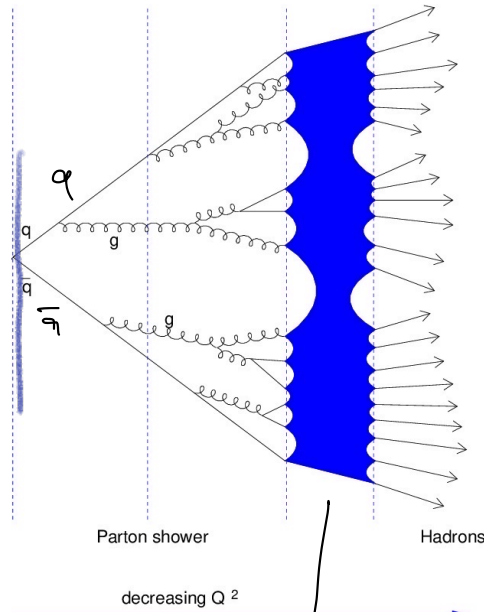
t-Channel.

$$e^+e^- \rightarrow e^+e^-$$

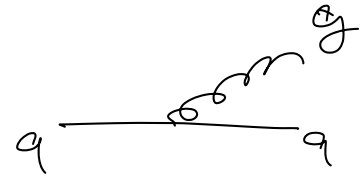
Bhabha scattering

$$\sqrt{s} = 500 \text{ MeV}$$

$$e^+e^- \rightarrow e^+e^-, \mu^+\mu^-, \pi^+\pi^-, \text{hadrons.}$$

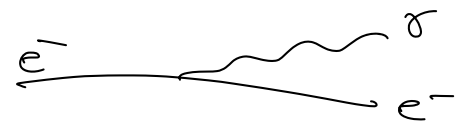


$$Q^2 \simeq \Lambda_{\text{QCD}}^2 \simeq 200 \text{ MeV}$$



gluonstrahlung

QCD equivalent of



Bremsstrahlung

gluon-parton splitting

if $\sqrt{s} \simeq 2m_H$

$$\sqrt{s} = 2m_\pi \Rightarrow e^+e^- \rightarrow \pi^+\pi^-$$

$$\sqrt{s} \simeq 2m_B = 10.580 \text{ GeV}$$

B-Factory

$$e^+e^- \rightarrow B^0 \bar{B}^0, B^+ B^-$$

S hadrons.
C hadrons.

